

ORIGINAL CONTRIBUTION

Independent Association of RNF213 p.R4810K With Extracranial Carotid Artery Disease Beyond Intracranial Atherosclerosis

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BACKGROUND: The *RNF213* p.R4810K variant, a susceptibility variant for Moyamoya disease, has also been linked to intracranial atherosclerotic disease (ICAD). However, its association with extracranial carotid artery disease (ECAD), particularly whether this association is independent of concomitant ICAD and whether it differs by stenosis severity, remains unclear.

METHODS: We conducted a multicenter registry-based observational study of patients with acute ischemic stroke (AIS) or transient ischemic attack enrolled in the Fukuoka Stroke Registry. Patients were genotyped for *RNF213* p.R4810K and classified as noncarriers (G/G) or variant carriers (G/A or A/A). ECAD was defined as $\geq 50\%$ stenosis or occlusion of the extracranial carotid arteries. Multivariable logistic regression was used to assess the association between *RNF213* p.R4810K and ECAD after adjusting for vascular risk factors and ICAD, and stratified analyses were performed by ICAD status. ECAD severity was modeled ordinally (none, moderate, severe, and occlusion) using a partial proportional-odds model.

RESULTS: Of 15 569 registry patients with AIS or transient ischemic attack, 13 994 were included in the present analysis after exclusions (mean age, 73 ± 12 years; 59.5% men). The *RNF213* p.R4810K variant was identified in 359 patients (2.6%). ECAD was present in 2181 patients (15.6%), including 78 variant carriers. Variant carriers had significantly higher odds of ECAD after adjustment for vascular risk factors and ICAD (odds ratio, 1.69 [95% CI, 1.29–2.22]). This association was consistent irrespective of ICAD status ($P_{\text{heterogeneity}} = 0.32$). In severity analyses, the association strengthened stepwise with increasing ECAD severity; the largest effect estimate was for occlusion (odds ratio, 4.55 [95% CI, 2.81–7.38]).

CONCLUSIONS: The *RNF213* p.R4810K variant was associated with ECAD independent of concomitant ICAD and showed a graded association with ECAD severity. Extracranial disease expression may involve modifiers distinct from those driving ICAD.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: genetics ■ hypertension ■ intracranial arteriosclerosis ■ ischemic stroke ■ risk factors

Acute ischemic stroke (AIS) is a leading cause of disability and mortality, and the underlying mechanisms are not fully understood. Large-artery atherosclerosis is a major cause of AIS and can arise from either extracranial carotid artery disease (ECAD) or intracranial

atherosclerotic disease (ICAD). In particular, ECAD is clinically important because stenosis severity directly influences the risk of recurrent ischemic events and guides decisions about carotid revascularization.^{1,2} Beyond traditional vascular risk factors such as hypertension and

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.126.055878>.

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Nonstandard Abbreviations and Acronyms

BMI	body mass index
ECAD	extracranial carotid artery disease
ECST	European Carotid Surgery Trial
ICAD	intracranial atherosclerotic disease
MMD	Moyamoya disease
RNF213	ring finger protein 213
TIA	transient ischemic attack
TOAST	Trial of ORG 10172 in Acute Stroke Treatment

dyslipidemia, recent studies have identified genetic factors that contribute to large-artery atherosclerotic disease and ischemic stroke.³

The *RNF213* (ring finger protein 213 gene, myosin) was originally identified as a susceptibility gene for Moyamoya disease (MMD), and the *RNF213* p.R4810K (rs112735431, c.14429G>A) variant is also increasingly recognized as an important genetic contributor to ICAD.^{4–6} In a large multicenter stroke registry in Japan, we recently reported that *RNF213* p.R4810K is associated with ICAD, with a stronger association observed among obese variant carriers.⁷ Notably, p.R4810K is relatively common in East Asians, whereas only a minority of carriers develop clinically overt vascular disease, supporting the two-hit hypothesis in which additional genetic or environmental factors are required for disease expression.^{8,9} Beyond intracranial disease, accumulating evidence has linked this variant to vascular abnormalities in extracranial arterial beds, including extracranial cervical artery abnormalities, coronary artery disease, and pulmonary arterial hypertension, leading to the broader concept of *RNF213* vasculopathy.^{10–14}

Although *RNF213* p.R4810K is increasingly established as a genetic risk factor for ICAD, its relevance to ECAD remains uncertain. Available evidence about extracranial carotid involvement has been limited and not entirely consistent: an earlier small subgroup analysis found no significant association with extracranial carotid atherosclerosis,¹⁵ whereas more recent studies have suggested a possible association between the variant and ECAD.^{12,13} Moreover, previous studies have not evaluated ECAD as a primary outcome; instead, extracranial carotid involvement has often been assessed only as a secondary or subgroup finding in studies primarily focused on intracranial disease. Consequently, a key clinical question remains unresolved: because ICAD and ECAD frequently coexist, any observed association with ECAD could be interpreted as secondary to the variant's established effects on intracranial arteries. Whether *RNF213* p.R4810K confers susceptibility to ECAD independently of coexisting ICAD, therefore, remains to be

determined. In addition, it has not been systematically examined whether the association, if present, varies across the severity spectrum of ECAD, including severe stenosis and occlusion.

Therefore, we evaluated the association between *RNF213* p.R4810K and ECAD in a large multicenter registry of patients with AIS and transient ischemic attack (TIA), focusing on whether the association is independent of concomitant ICAD and whether it differs across the spectrum of ECAD stenosis severity.

METHODS

Data Availability

Anonymized data will be shared upon request from any qualified investigator for the purpose of replicating the study procedures and results. Requests should be directed to the corresponding author.

Ethics Statement

The study design was approved by the institutional review boards of all of the following participating hospitals: Kyushu University Institutional Review Board for Clinical Research, 22086-00; Kyushu Medical Center Institutional Review Board, R06-03; Clinical Research Review Board of Fukuoka Higashi Medical Center, 29-C-38; Fukuoka Red Cross Hospital Institutional Review Board, 629; St. Mary's Hospital Research Ethics Review Committee, S13-0110; Steel Memorial Yawata Hospital Ethics Committee, 06-04-13; and Kyushu Rosai Hospital institutional review board, 21-8. Written informed consent was obtained from all patients or surrogate family members. This study was performed in accordance with the Declaration of Helsinki and its subsequent amendments.

Study Subjects

Study participants were identified from the Fukuoka Stroke Registry, a prospective, multicenter, hospital-based registry of consecutive patients with acute stroke or TIA admitted to 7 stroke centers in Fukuoka, Japan, within 7 days of symptom onset (<https://www.umin.ac.jp/ctr>; unique identifier: 000000800). This study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology Guidelines.¹⁶ The registry design, methods of data collection, and harmonization of the Fukuoka Stroke Registry have been described previously.^{17,18}

Stroke was defined as a sudden onset of a nonconvulsive neurological deficit persisting for ≥ 24 hours. Stroke subtype was classified based on clinical findings and brain imaging, including computed tomography and magnetic resonance imaging. Ischemic stroke was diagnosed by stroke neurologists based on clinical presentation, brain and vascular imaging findings, and the subsequent clinical course. TIA was defined as a focal cerebral ischemic event with neurological symptoms lasting < 24 hours, with no brain lesions detected by brain computed tomography or magnetic resonance imaging. Diagnosis was confirmed upon admission by at least 2 stroke neurologists.

Genotyping

Blood samples were drawn at enrollment after patients had provided informed consent for genetic analysis. The samples were stored at -80°C before DNA extraction, and total genomic DNA was extracted from whole blood using a standardized protocol (SRL, Tokyo, Japan) and stored at -80°C until use. The *RNF213* c.14429G>A (rs112735431, p.R4810K) variant was screened using Melt Analysis of Mismatch Amplification Mutation Assays based on fluorescent melting curve analysis using real-time polymerase chain reaction as previously described^{7,19} and confirmed by Sanger sequencing. Real-time polymerase chain reaction was performed on a LightCycler 96 (Roche Diagnostics GmbH). Each 10- μL reaction contained 5- μL KOD SYBR qPCR Mix (Toyobo), 0.5- $\mu\text{mol/L}$ primers, and 10 to 25 ng of genomic DNA. The polymerase chain reaction conditions were 98°C for 2 minutes, followed by 35 cycles of 98°C for 10 s, 60°C for 10 s, and 68°C for 30 s. After amplification, polymerase chain reaction products underwent melting curve analysis using a dissociation protocol of 95°C for 10 s followed by an incremental temperature ramp (0.2°C increments) from 65°C to 97°C . SYBR Green fluorescence intensity was measured at 530 nm at each ramp interval and plotted against temperature. The following primers were used (Sigma-Aldrich, Tokyo, Japan): *RNF213*^{wt(14429G)} forward primer (5'-aaGCAAGTTGAATACAGCTCCATgAG-3'); *RNF213*^{R4810K(14429G>A)} forward primer (5'-gccgcctgcccgcGCAAGTTGAATACAGCTCCATtAA-3'); and a common reverse primer (5'-AGCCACACCTGAGCTGTGCTT-3'). The noncomplementary nucleotides are written in lowercase. Patients with heterozygous GA or homozygous AA genotypes were classified as p.R4810K variant carriers. Independent investigators blinded to patient identity and phenotype assigned genotypes to the database.

Definitions of ECAD, ICAD, MMD, and Cerebral Artery Dissection

ECAD was defined as 50% to 99% stenosis or occlusion of the carotid artery as measured by a standardized method.¹ This definition included lesions of the common carotid arteries and cervical internal carotid arteries, which were assessed using carotid duplex ultrasonography. Stenosis severity was classified according to the ECST (European Carotid Surgery Trial) criteria as follows: <50% as none, 50% to 69% as moderate, 70% to 99% as severe, and complete occlusion.^{1,20,21} When reliable ECST measurement was technically difficult (eg, marked calcification or poor visualization), Doppler velocity criteria were applied. A peak systolic velocity ≥ 125 cm/s, as defined by the Society of Radiologists in Ultrasound consensus,²¹ was considered equivalent to NASCET $\geq 50\%$ stenosis,² which corresponds to $\approx 70\%$ stenosis in ECST terms based on the conversion formula ($\text{ECST}\% = 0.6 \times \text{NASCET}\% + 40$).²² For patients with a history of carotid endarterectomy or carotid artery stenting, the treated artery was classified as having severe ECAD and was assigned to the severe stenosis category for the severity analyses (65 patients; 61 noncarriers and 4 variant carriers). ICAD, MMD, and cerebral artery dissection were diagnosed based on intracranial vascular imaging, including computed tomography angiography, MR angiography, and digital subtraction angiography. ICAD was defined as 50% to 99% stenosis or occlusion of the intracranial artery as measured by a standardized method,

as reported previously.^{7,23} Acute large-vessel occlusions clinically diagnosed as embolic were excluded from the ECAD and ICAD categories (1400 patients, including 1366 noncarriers and 34 *RNF213* p.R4810K carriers). Occlusion was categorized as ECAD/ICAD only when underlying atherosclerotic stenosis or chronic occlusion was documented on follow-up or postrecanalization imaging. MMD was defined according to the diagnostic criteria of the Moyamoya Disease Study Group of the Ministry of Health, Labour, and Welfare,²⁴ and extracranial and intracranial arterial dissection was defined according to the diagnostic criteria of the European Stroke Organization 2021.²⁵ All imaging studies used for case definition were initially interpreted at each participating hospital by at least 1 radiologist and 1 stroke neurologist, and ambiguous cases were adjudicated through clinical review by stroke neurologists. For ECAD and ICAD assessments, vascular imaging assessors were blinded to the *RNF213* p.R4810K genotype.

Clinical Assessment

Body mass index (BMI) was calculated as body weight in kilograms divided by height in meters squared. Hypertension was considered present when blood pressure was $\geq 140/90$ mm Hg in the chronic stage or when there was a medical history of hypertension. Diabetes was determined according to the criteria of the Japan Diabetes Society, either before stroke onset or during hospitalization.²⁶ Dyslipidemia was defined as a low-density lipoprotein cholesterol concentration of ≥ 140 mg/dL, a high-density lipoprotein cholesterol concentration of < 40 mg/dL, or a triglyceride concentration of ≥ 150 mg/dL during hospitalization or in the chronic stage, or as prestroke treatment with antihyperlipidemic agents. The estimated glomerular filtration rate was estimated using the equation recommended by the Japanese Society of Nephrology: estimated glomerular filtration rate ($\text{mL/min per } 1.73 \text{ m}^2$) = $194 \times \text{serum creatinine (mg/dL)}^{-1.094} \times \text{age (years)}^{-0.287} \times 0.739$ (if female).²⁷ Atrial fibrillation was identified from previous history or electrocardiography during hospitalization. Information on smoking habits (previous or current cigarette smoking) and alcohol intake was collected using a standardized questionnaire. Previous stroke was defined as a history of hemorrhagic or ischemic stroke before the index stroke. Coronary artery disease was defined as a history of angina pectoris, myocardial infarction, percutaneous coronary intervention, or coronary artery bypass graft surgery. Information on family history of stroke in first-degree relatives, including parents, children, and siblings, was collected. Ischemic stroke was further classified into 4 subtypes: lacunar infarction, atherothrombotic infarction, cardioembolic stroke, and unclassified infarction, according to the diagnostic criteria of the TOAST study (Trial of ORG 10172 in Acute Stroke Treatment).²⁸ The medical history and risk factors of patients were documented at the time of enrollment.

Statistical Analysis

Baseline characteristics were compared among genotypes using the 2-sided unpaired Student *t* test for continuous variables and the χ^2 test for categorical variables. The association between the *RNF213* genotypes and ECAD was assessed using logistic regression analysis. Logistic regression analysis was performed to estimate the odds ratio (OR) and 95% CI of each outcome of interest after adjusting for covariates.

We used a 3-step approach for covariate adjustment: model 1 was adjusted for age and sex; model 2 was adjusted for the variables included in model 1 and BMI, hypertension, diabetes, dyslipidemia, smoking, and drinking; and model 3 was further adjusted for the variables included in model 2 and ICAD to examine whether the association between the *RNF213* p.R4810K variant and ECAD persisted after accounting for concomitant ICAD. To verify that the association between the *RNF213* p.R4810K variant and ECAD was independent of the association between the *RNF213* p.R4810K variant and ICAD, we performed stratified multivariable logistic regression by ICAD status. Within each stratum, models were adjusted for age, sex, BMI, hypertension, diabetes, dyslipidemia, smoking, and drinking. In addition, exploratory subgroup analyses were conducted according to age (<70/≥70 years), sex, BMI (<30/≥30 kg/m²), hypertension, diabetes, dyslipidemia, smoking, and drinking using multivariable logistic regression corresponding to model 2 (excluding the stratification variable). The heterogeneity in the association between the *RNF213* p.R4810K variant and ECAD with respect to covariates was also evaluated by including a multiplicative interaction term in the logistic regression model. In addition, to address potential effect modification after accounting for concomitant ICAD, we examined multiplicative interaction terms between *RNF213* p.R4810K carrier status and each covariate in model 3, with each interaction term entered separately. Because ECAD severity is ordinal (none, moderate, severe, and occlusion), we next modeled severity using an ordinal logistic regression with cumulative logits. The proportional-odds assumption was assessed by the Brant test.²⁹ Given evidence of nonproportionality, we fitted a partial proportional-odds model that relaxed the proportional-odds assumption only for the *RNF213* p.R4810K variant and retained it for all other covariates, and we report threshold-specific adjusted ORs (≥moderate versus none; ≥severe versus ≤moderate; and occlusion versus ≤severe). As a sensitivity analysis, we also fitted a multinomial logistic regression treating the 4 severity levels as nominal categories (reference: none) and present relative risk ratios. Among the covariates included in the multivariable models, only BMI had missing data (n=6); therefore, complete-case analysis was performed without imputation.

Two-tailed $P < 0.05$ was considered statistically significant. Statistical analyses were performed using Stata version 16.0 (StataCorp, College Station, TX).

RESULTS

Study Population

A total of 15 569 patients with AIS or TIA were included in a prospective Fukuoka Stroke Registry study, from June 2007 to September 2019. After excluding 347 patients who did not consent to our genomic study, 749 patients without imaging modalities of carotid duplex ultrasonography, 197 patients without intracranial vascular imaging, 33 patients who were diagnosed with MMD, and 249 patients with acute cerebral and cervical artery dissection, the final study group included 13 994 patients (Figure 1). Complete vascular assessment, including carotid duplex ultrasonography for ECAD assessment and intracranial vascular imaging for ICAD assessment, was available in 14 276 of the 15 222 patients who consented to the genomic study (93.8%).

Among patients retained in the analysis, possible overlap with a quasi-moyamoya phenotype was identified in only 3 patients, including 1 *RNF213* p.R4810K carrier; none had been clinically diagnosed with MMD.

Baseline Characteristics According to *RNF213* p.R4810K Carrier Status

The mean (SD) age of the 13 994 patients was 73 (±12) years, and 8333 (59.5%) were men. Overall, 13 635 (97.4%) were noncarriers, and 359 (2.6%) were carriers of the *RNF213* p.R4810K variant. Baseline characteristics according to p.R4810K carrier status are shown in Table 1. Compared with noncarriers, carriers were younger and had a higher mean BMI. Carriers more frequently had atherothrombotic stroke, TIA, and ICAD, whereas lacunar stroke and unclassified stroke were less frequent.

Association Between the *RNF213* p.R4810K Variant and ECAD

ECAD was identified in 2181 patients (15.6%): 2103 noncarriers and 78 carriers. Table 2 shows ORs and 95%

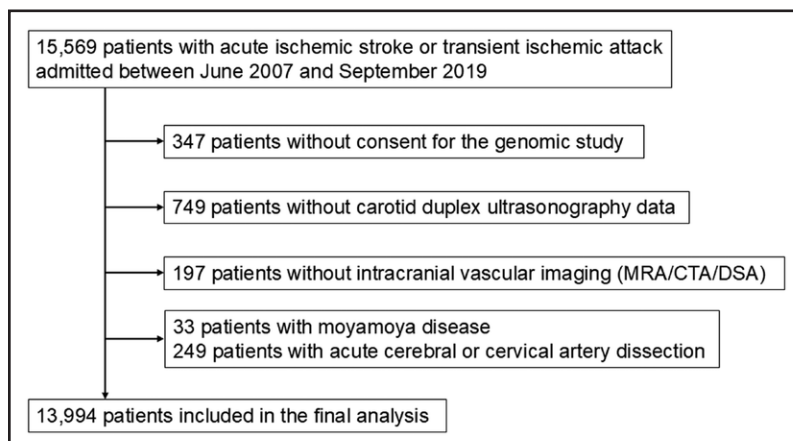


Figure 1. Flowchart of patient selection for the main analysis.

CTA indicates computed tomography angiography; DSA, digital subtraction angiography; and MRA, magnetic resonance angiography.

Table 1. Baseline Characteristics of the *RNF213* p.R4810K Variant Carriers and Noncarriers

Characteristics	Noncarriers (n=13 635)	Carriers (n=359)	P value
Age, y; mean (SD)	73 (12)	70 (13)	<0.001
Men, n (%)	8124 (59.6)	209 (58.2)	0.60
BMI, kg/m ² ; mean (SD)	22.9 (3.8)	23.6 (4.3)	<0.01
Hypertension, n (%)	11 089 (81.3)	306 (85.2)	0.06
Diabetes, n (%)	4259 (31.2)	111 (30.9)	0.90
Dyslipidemia, n (%)	7603 (55.8)	207 (57.7)	0.47
eGFR, mL/(min·1.73 m ²); mean (SD)	66.1 (24.7)	66.1 (26.8)	0.99
Atrial fibrillation, n (%)	3249 (23.8)	76 (21.2)	0.24
Smoking, n (%)	7311 (53.6)	195 (54.3)	0.79
Drinking, n (%)	4662 (34.2)	129 (35.9)	0.49
Previous stroke or TIA, n (%)	2491 (18.3)	61 (17.0)	0.53
Coronary artery disease, n (%)	1981 (14.5)	61 (17.0)	0.19
Family history of stroke, n (%)	4997 (37.7)	134 (37.9)	0.97
Ischemic stroke subtype, n (%)			
Lacunar	2543 (18.7)	42 (11.7)	<0.001
Atherothrombotic	2224 (16.3)	105 (29.2)	<0.001
Cardioembolic	2838 (20.8)	69 (19.2)	0.46
Unclassified	5010 (36.8)	98 (27.3)	<0.001
TIA, n (%)	1013 (7.4)	45 (12.5)	<0.001
ICAD, n (%)	3103 (22.8)	150 (41.8)	<0.001

BMI indicates body mass index; eGFR, estimated glomerular filtration rate; and TIA, transient ischemic attack.

CIIs for the association between p.R4810K carrier status and the presence of ECAD. The OR adjusted for age and sex was significantly higher in the *RNF213* p.R4810K carriers (model 1), and this association remained significant after adjusting for other confounding factors (model 2). Furthermore, adjusting for the presence of ICAD did not substantially reduce the OR, which remained significantly higher in carriers (model 3).

Independence From ICAD: Stratified Analyses by ICAD

To assess whether the association between the *RNF213* p.R4810K variant and ECAD was independent of coexisting ICAD, we performed subgroup analyses stratified by ICAD status. When stratified by ICAD status, the adjusted associations were similar (ICAD−: OR, 1.50 [95% CI, 1.04–2.19]; ICAD+: OR, 1.78 [95% CI, 1.20–2.65]), with no significant heterogeneity across

ICAD status ($P_{\text{heterogeneity}}=0.32$; Figure 2). In contrast to our prior ICAD analysis, exploratory subgroup analyses showed no evidence of statistically significant heterogeneity across subgroups, including age and BMI categories and other conventional vascular risk factors (Table S1). Similarly, when multiplicative interaction terms between *RNF213* p.R4810K carrier status and each covariate were examined in model 3 additionally adjusted for ICAD, no statistically significant interactions were observed (Table S2).

Association Between *RNF213* p.R4810K Variant and ECAD Severity

The crude distribution of ECAD severity by p.R4810K carrier status is shown in Table S3. The proportion of carriers increased across more severe ECAD categories, particularly severe stenosis and occlusion. Because ECAD severity was classified as none, moderate, severe,

Table 2. Association Between the *RNF213* p.R4810K Variant and ECAD

	No. of	Model 1		Model 2		Model 3	
	events (%)	OR (95% CI)	P value	OR (95% CI)	P value	OR (95% CI)	P value
ECAD							
Noncarriers	2103 (15.4)	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Carriers	78 (21.7)	1.79 (1.37–2.34)	<0.001	1.80 (1.37–2.36)	<0.001	1.69 (1.29–2.22)	<0.001

Model 1 was adjusted for age and sex. Model 2 was adjusted for the variables included in model 1 and BMI, hypertension, diabetes, dyslipidemia, smoking, and drinking. Model 3 was adjusted for the variables included in model 2 and ICAD. BMI indicates body mass index; ECAD, extracranial carotid artery disease; ICAD, intracranial atherosclerotic disease; and OR, odds ratio.

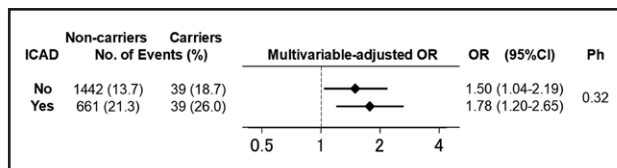


Figure 2. Association between *RNF213* p.R4810K and extracranial carotid artery disease stratified by intracranial atherosclerotic disease (ICAD) status.

The multivariable model was adjusted for age, sex, body mass index, hypertension, diabetes, dyslipidemia, smoking, and drinking. OR indicates odds ratio; and Ph, *P* value for heterogeneity.

or occlusion, we modeled severity using an ordinal logistic regression model with cumulative logits. The Brant test indicated a violation of the proportional-odds assumption for the p.R4810K term (Table S4); therefore, we fitted a partial proportional-odds model that relaxed the proportional-odds assumption for p.R4810K while constraining all other covariates. In this model, adjusted ORs for carriers versus noncarriers increased with ECAD severity (Table 3): $\geq$ moderate versus none, 1.76 (95% CI, 1.34–2.31); $\geq$ severe versus $\leq$ moderate, 2.25 (95% CI, 1.65–3.06); and occlusion versus $\leq$ severe, 4.55 (95% CI, 2.81–7.38; all $P < 0.001$), suggesting a graded association. In a sensitivity analysis using multinomial logistic regression treating ECAD severity as nominal categories, the pattern of associations was similar (Table S5).

DISCUSSION

In this large stroke/TIA-based cohort, the *RNF213* p.R4810K variant was independently associated with ECAD after adjustment for conventional vascular risk factors and ICAD. Effect estimates were similar regardless of ICAD status, indicating that the higher odds of ECAD in carriers are not explained solely by coexisting intracranial disease. Moreover, the association between *RNF213* p.R4810K and ECAD was progressively stronger with increasing ECAD severity, suggesting a severity-dependent, graded association across ECAD categories.

Recent evidence has shown that the *RNF213* p.R4810K variant is associated with various vascular diseases beyond MMD, leading to the concept of *RNF213* vasculopathy. In addition to its well-established association with ICAD, this variant has also been linked

to extracranial cervical artery abnormalities, coronary artery disease, and pulmonary arterial hypertension.^{10–14} In this context, our finding that *RNF213* p.R4810K was associated with ECAD is consistent with the emerging view that *RNF213*-related vascular vulnerability is not confined to the intracranial circulation. However, previous studies addressing extracranial arterial involvement have generally been limited by small sample sizes or hypothesis-generating exploratory analyses, with mixed findings about extracranial carotid atherosclerosis.^{12,13,15} The present study extends the phenotypic spectrum of *RNF213* vasculopathy by providing evidence of an association between the variant and ECAD in a large multi-center registry of patients with AIS and TIA.

ECAD and ICAD are both considered manifestations of atherosclerotic large-artery disease and frequently coexist in clinical practice. Accordingly, it might be assumed that the association between *RNF213* p.R4810K and ECAD is secondary to its established effects on intracranial arteries. However, in the present study, adjustment for ICAD did not materially attenuate the association between the variant and ECAD (Table 2), and the subgroup analysis stratified by ICAD status showed similar associations in patients with and without ICAD (Figure 2). These findings suggest that *RNF213* p.R4810K is associated with higher odds of ECAD even after accounting for ICAD. Accordingly, our results support the view that *RNF213*-related vasculopathy should be regarded as a systemic large-artery disorder rather than a disease confined to the intracranial circulation. They also highlight the need to identify determinants that govern whether the vascular phenotype manifests predominantly as ECAD or ICAD.

From a mechanistic standpoint, these epidemiological findings may be interpreted in the context of the two-hit hypothesis for *RNF213*-related vasculopathy.^{8,9,30} The *RNF213* p.R4810K variant is relatively common in East Asian populations, yet only a minority of carriers develop clinically overt vascular disease, suggesting that additional genetic or environmental insults are required for disease expression. Experimental studies have demonstrated that *RNF213* and the p.R4810K variant influence angiogenesis and link inflammatory and angiogenic signaling in endothelial cells.^{31–33} In addition, *RNF213* has been proposed to interact with pathways related to

Table 3. Threshold-Specific Adjusted ORs for ECAD Severity by *RNF213* p.R4810K Carrier Status (Partial Proportional-Odds Model)

Outcome threshold	Model 1		Model 2	
	OR (95% CI)	<i>P</i> value	OR (95% CI)	<i>P</i> value
$\geq$ Moderate vs none	1.75 (1.34–2.28)	<0.001	1.76 (1.34–2.31)	<0.001
$\geq$ Severe vs $\leq$ moderate	2.23 (1.65–3.03)	<0.001	2.25 (1.65–3.06)	<0.001
Occlusion vs $\leq$ severe	4.53 (2.82–7.27)	<0.001	4.55 (2.81–7.38)	<0.001

Model 1 was adjusted for age and sex. Model 2 was adjusted for the variables included in model 1 and BMI, hypertension, diabetes, dyslipidemia, smoking, and drinking. BMI indicates body mass index; ECAD, extracranial carotid artery disease; and OR, odds ratio.

inflammation and metabolic traits such as obesity and insulin resistance.^{30,34} In our previous report on ICAD, we observed significant heterogeneity in the association between *RNF213* p.R4810K and ICAD according to BMI and age; in particular, higher BMI (ie, obesity) appeared to act as a potential second hit that amplified the effect of *RNF213* p.R4810K on ICAD risk.⁷ In the present study, however, given the limited number of *RNF213* p.R4810K carriers with ECAD and the need for cautious interpretation, we did not observe significant effect modification of the RNF213–ECAD association by BMI, age, or other conventional vascular risk factors (Table S1). The contrast with our previous ICAD findings suggests that the nature of the second hit required for RNF213-related vasculopathy may differ between intracranial and extracranial carotid arterial beds. One possibility is that the relevant second hits for RNF213-related vasculopathy are vascular bed-specific. Differences in arterial wall architecture along the extracranial–intracranial arterial continuum, including the muscular-to-elastic transition characteristics of extracranial carotid arteries and the distinct development of adventitial vasa vasorum compared with intracranial arteries,^{35,36} and local microenvironmental stresses (hemodynamic load, hypoxia, and immune-endothelial interactions)^{37–39} may modify how RNF213-related susceptibility translates into ECAD versus ICAD. Collectively, these observations raise the hypothesis that *RNF213* p.R4810K provides a shared baseline susceptibility, whereas vascular bed-specific second hits may help determine whether the phenotype manifests predominantly as ECAD or ICAD.

Another important finding was that the association between *RNF213* p.R4810K and ECAD became progressively stronger with increasing stenosis severity, with the largest effect estimate observed for occlusion. This severity-dependent pattern was consistent in both the partial proportional-odds model and a multinomial logistic regression sensitivity analysis (Table 3; Table S5), suggesting that, in this AIS/TIA-based cohort, effect estimates were larger for more severe ECAD categories than for mild-to-moderate disease. One possible explanation is that RNF213-related vascular vulnerability may influence the mode of arterial remodeling, potentially favoring inward (negative) remodeling in which luminal narrowing may occur disproportionately to plaque burden, thereby contributing to more severe stenosis or occlusion.^{40,41} This pattern is also compatible with the possibility that progressive luminal narrowing and hemodynamic compromise, rather than plaque disruption and embolization, play a relatively larger role in variant carriers, although this remains speculative and warrants further study. Alternatively, because the present study is based on patients with AIS and TIA, the observed severity-dependent association may partly reflect clinical ascertainment or Berkson's bias. Variant carriers may be more likely to be captured once ECAD has progressed to a stage that is more likely to

contribute to AIS/TIA, whereas milder ECAD may be less likely to be represented in a stroke/TIA-based registry. Therefore, the severity-dependent association observed in this study should be interpreted in the context of an AIS/TIA registry and should not be directly extrapolated to the general population of *RNF213* p.R4810K carriers.

This study has several strengths and limitations. A major strength is the large sample size (13 994 patients), with high rates of genetic testing (97.8%) and vascular imaging (93.8%), as well as limited missing data, which allowed us to perform detailed subgroup analyses and to evaluate ECAD according to stenosis severity. By contrast, the present study included patients who developed AIS or TIA, regardless of whether ECAD was symptomatic or asymptomatic, introducing a potential selection bias. In addition, ECAD was assessed uniformly using standardized carotid duplex ultrasonography, which is operator-dependent and has inherent limitations compared with CTA or DSA. However, the use of a single non-invasive modality across the cohort minimized differential ascertainment that could have arisen from selectively performed contrast-based imaging. The study population was restricted to a specific region in Japan, potentially limiting the generalizability of the findings. Nevertheless, the epidemiological relevance of the *RNF213* p.R4810K variant can only be adequately assessed in populations with a high prevalence of this variant, such as East Asians, whereas its functional impact at the molecular level is thought to be consistent across different ethnic backgrounds. Finally, because of the cross-sectional design, the timing of ECAD onset could not be determined, highlighting the need for future longitudinal research in populations less subject to this selection bias, such as community-based or health-screening cohorts. These studies should use standardized vascular imaging to evaluate ICAD and ECAD severity, lesion distribution, disease progression, and subsequent cerebrovascular or cardiovascular events in *RNF213* p.R4810K carriers.

CONCLUSIONS

In this multicenter stroke/TIA registry, the *RNF213* p.R4810K variant was independently associated with ECAD, with generally consistent effects regardless of ICAD status and a stronger association at higher disease severity. This vascular bed-specific association of the *RNF213* p.R4810K variant supports the possibility of distinct mechanisms underlying extracranial versus intracranial disease expression. Future studies are warranted to identify modifiers that shape extracranial disease expression.

ARTICLE INFORMATION

Received March 11, 2026; final revision received July 15, 2026; accepted August 3, 2026.

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Acknowledgments

The authors thank all study participants, the Fukuoka Stroke Registry investigators for collecting data, and all clinical research coordinators (Hisayama Research Institute for Lifestyle Diseases) for their help in obtaining informed consent and collecting clinical data.

Sources of Funding

This study was supported by the Japan Society for the Promotion of Science, KAKENHI (grants JP21K10330, JP21K19648, JP21H03165, and JP24K02669); the Astellas Foundation for Research on Metabolic Disorders; and Grant of the Clinical Research Promotion Foundation (2023).

Disclosures

None.

Supplemental Material

Tables S1–S5

Supplemental Appendix

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